CHEMISTRY

SECTION-A

51. Match List I with List II.

List-I (Process)		List-II (Conditions)	
A.	Isothermal process	I.	No heat exchange
B.	Isochoric process	II.	Carried out at constant temperature
C.	Isobaric process	III.	Carried out at constant volume
D.	Adiabatic process	IV.	Carried out at constant pressure

Choose the correct answer from the options given below:

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-III, C-IV, D-I

(3) A-IV, B-III, C-II, D-I

(4) A-IV, B-II, C-III, D-I

Answer (2)

Sol. (A) Isothermal process $\Rightarrow$ Temperature is constant throughout the process

(B) Isochoric process $\Rightarrow$ Volume is constant throughout the process

(C) Isobaric process $\Rightarrow$ Pressure is constant throughout the process

(D) Adiabatic process $\Rightarrow$ No exchange of heat (q) between system and surrounding

52. Given below are two statements:

Statement I : The boiling point of three isomeric pentanes follows the order

n-pentane > isopentane > neopentane

Statement II : When branching increases, the molecule attains a shape of sphere. This results in smaller surface area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are correct

(4) Both Statement I and Statement II are incorrect

Answer (3)

Sol. Both statement I and statement II are correct.

Boiling point of n-pentane = 309 K

isopentane = 301 K

neopentane = 282.5

As branching increases molecules attain the shape of a sphere results in smaller area of contact thus weak intermolecular forces between spherical molecules, which are overcome at relatively lower temperature. Leading to decrease in boiling point.

53. Match List I with List II.

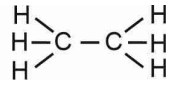
	List I (Molecule)		List II (Number and types of bond/s between two carbon atoms)
A.	ethane	I.	one σ-bond and two π-bonds
B.	ethene	II.	two π-bonds
C.	carbon molecule, C_2	III.	one σ-bond
D.	ethyne	IV.	one σ-bond and one π-bond

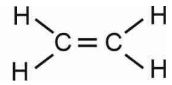
Choose the correct answer from the options given below:

(1) A-III, B-IV, C-II, D-I (2) A-III, B-IV, C-I, D-II

(3) A-I, B-IV, C-II, D-III (4) A-IV, B-III, C-II, D-I

Answer (1)

Sol. (A) Ethane  one (C – C) σ bond

(B) Ethene  one (C – C) σ and one (C – C) π bond

(C) C_2 two (C – C) π bonds

(D) Ethyne H – C ≡ C – H two (C – C) π bonds and one (C – C) σ bond

54. Which one of the following alcohols reacts instantaneously with Lucas reagent?

(1) $CH_3 - CH(CH_3) - CH_2OH$

(2) $CH_3 - C(CH_3)_2 - OH$

(3) $CH_3 - CH_2 - CH_2 - CH_2OH$

(4) $CH_3 - CH_2 - CH(CH_3) - OH$

Answer (2)

Sol. Tertiary alcohols react instantaneously with Lucas reagent and gives immediate turbidity.

In case of tertiary alcohols, they form halides easily with Lucas reagent (conc. HCl and $ZnCl_2$)

55. Match List I with List II

	List I (Quantum Number)		List II (Information provided)
A.	m_l	I.	Shape of orbital
B.	m_s	II.	Size of orbital
C.	l	III.	Orientation of orbital
D.	n	IV.	Orientation of spin of electron

Choose the correct answer from the options given below :

(1) A-III, B-IV, C-II, D-I

(2) A-II, B-I, C-IV, D-III

(3) A-I, B-III, C-II, D-IV

(4) A-III, B-IV, C-I, D-II

Answer (4)

Sol.
- Magnetic quantum number m_l informs about orientation of orbital.
- Spin quantum number m_s informs about orientation of spin of electron.
- Azimuthal quantum number (l) informs about shape of orbital
- Principal quantum number (n) informs about size of orbital

56. 'Spin only' magnetic moment is same for which of the following ions?

A. Ti^{3+}

B. Cr^{2+}

C. Mn^{2+}

D. Fe^{2+}

E. Sc^{3+}

Choose the most appropriate answer from the options given below.

(1) B and C only

(2) A and D only

(3) B and D only

(4) A and E only

Answer (3)

Sol.

Ions	No. of unpaired electrons	Configuration
Ti^{3+}	(3)	$3d^1$
Cr^{2+}	4	$3d^4$
Mn^{2+}	5	$3d^5$
Fe^{2+}	4	$3d^6$
Sc^{3+}	0	$3d^0$

Spin only magnetic moment is given by $\sqrt{n(n+2)}$BM

$\therefore$ Cr^{2+} and Fe^{2+} will have same spin only magnetic moment.

57. On heating, some solid substances change from solid to vapour state without passing through liquid state. The technique used for the purification of such solid substances based on the above principle is known as

(1) Distillation

(2) Chromatography

(3) Crystallization

(4) Sublimation

Answer (4)

Sol. (1) **Distillation :** It is used to separate volatile liquids from non-volatile impurities and the liquids having sufficient difference in their boiling point.

(2) **Chromatography :** It is based on separation by using stationary and mobile phase.

(3) **Crystallization :** It is based on difference in the solubilities of the compound and impurities in a suitable solvent.

(4) **Sublimation :** It is the purification technique based on principle that on heating, some solid substances change from solid to vapour state without passing through liquid state.

58. The energy of an electron in the ground state (n = 1) for He^+ ion is –x J, then that for an electron in n = 2 state for Be^{3+} ion in J is

(1) –4x

(2) $-\frac{4}{9}x$

(3) –x

(4) $-\frac{x}{9}$

Answer (3)

Sol. $E_n = -R_H\left(\frac{Z^2}{n^2}\right)J$

For He^+ (n = 1),

$$E_n = -x = -R_H\left(\frac{2^2}{1^2}\right) = -4R_H$$

$$\therefore \quad R_H = \frac{x}{4}$$

For Be^{3+} (n = 2),

$$E_n = -R_H\left(\frac{Z^2}{n^2}\right)J$$

$$= -\frac{x}{4}\times\left(\frac{4\times 4}{2\times 2}\right) = -x\ J$$

59. Activation energy of any chemical reaction can be calculated if one knows the value of

(1) orientation of reactant molecules during collision

(2) rate constant at two different temperatures

(3) rate constant at standard temperature

(4) probability of collision

Answer (2)

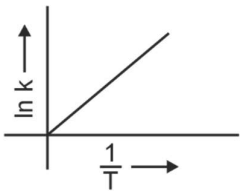
Sol. To calculate value of E_a

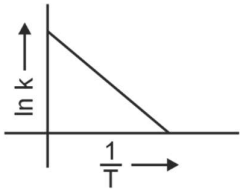
Equation used is

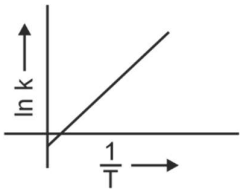
$$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

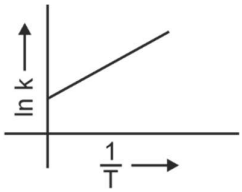
Hence E_a can be calculated if value of rate constant k is known at two different temperatures T_1 and T_2.

60. Which plot of ln k vs $\frac{1}{T}$ is consistent with Arrhenius equation?

(1) 

(2) 

(3) 

(4) 

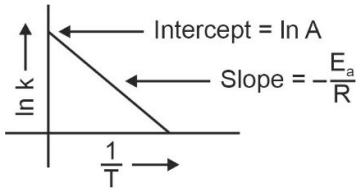
Answer (2)

Sol. The Arrhenius equation is given as

$$k = Ae^{-\frac{E_a}{RT}}$$

$$\therefore \quad \ln k = \ln A - \frac{E_a}{RT}$$

ln k v/s $\frac{1}{T}$ gives a straight line graph with slope = $-\frac{E_a}{R}$ and intercept = ln A



61. Given below are two statements:

Statement I: The boiling point of hydrides of Group 16 elements follow the order

$H_2O > H_2Te > H_2Se > H_2S$.

Statement II: On the basis of molecular mass, H_2O is expected to have lower boiling point than the other members of the group but due to the presence of extensive H-bonding in H_2O, it has higher boiling point.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

Answer (3)

Sol. **Statement I** is correct, because boiling point of hydrides of group 16 follows the order

$H_2O > H_2Te > H_2Se > H_2S$.

Statement II due to intermolecular H-bonding H_2O shows higher boiling point than respective hydrides of group 16.

(Both Statement are true)

Order from H_2Te to H_2S is due to decreasing molar mass.

62. The reagents with which glucose does **not** react to give the corresponding tests/products are

A. Tollen's reagent

B. Schiff's reagent

C. HCN

D. NH_2OH

E. $NaHSO_3$

Choose the correct options from the given below:

(1) B and E (2) E and D

(3) B and C (4) A and D

Answer (1)

Sol. Despite having the aldehyde group glucose does not give Schiff's test and it does not form the hydrogen sulphite addition product with $NaHSO_3$.

63. Arrange the following elements in increasing order of first ionization enthalpy:

Li, Be, B, C, N

Choose the correct answer from the options given below:

(1) Li < Be < C < B < N (2) Li < Be < N < B < C

(3) Li < Be < B < C < N (4) Li < B < Be < C < N

Answer (4)

Sol. Increasing order of first ionization enthalpy is Li < B < Be < C < N

Element	First ionization enthalpy (Δ_iH/kJ mol^{-1})
Li	520
Be	899
B	801
C	1086
N	1402

64. Arrange the following elements in increasing order of electronegativity:

N, O, F, C, Si

Choose the correct answer from the options given below:

(1) O < F < N < C < Si (2) F < O < N < C < Si

(3) Si < C < N < O < F (4) Si < C < O < N < F

Answer (3)

Sol. Electronegativity increases across the period on moving left to right. It decreases on moving down the group.

The correct option is Si < C < N < O < F

65. The E° value for the Mn^{3+}/Mn^{2+} couple is more positive than that of Cr^{3+}/Cr^{2+} or Fe^{3+}/Fe^{2+} due to change of

(1) d^4 to d^5 configuration

(2) d^3 to d^5 configuration

(3) d^5 to d^4 configuration

(4) d^5 to d^2 configuration

Answer (1)

Sol. $E^{\circ}_{Mn^{3+}/Mn^{2+}} > E^{\circ}_{Cr^{3+}/Cr^{2+}}$ or $E^{\circ}_{Fe^{3+}/Fe^{2+}}$

Electronic configuration of Mn^{3+} = [Ar]3d^4

Electronic configuration of Mn^{2+} = [Ar]3d^5

Electronic configuration of Cr^{3+} = [Ar]3d^3

Electronic configuration of Cr^{2+} = [Ar]3d^4

As Mn^{3+} from d^4 configuration goes to more stable d^5 configuration (Half filled), due to more exchange energy in d^5 configuration.

66. Match List I with List II.

	List I (Compound)		**List II (Shape/geometry)**
A.	NH_3	I.	Trigonal Pyramidal
B.	BrF_5	II.	Square Planar
C.	XeF_4	III.	Octahedral
D.	SF_6	IV.	Square Pyramidal

Choose the correct answer from the options given below:

(1) A-III, B-IV, C-I, D-II

(2) A-II, B-III, C-IV, D-I

(3) A-I, B-IV, C-II, D-III

(4) A-II, B-IV, C-III, D-I

Answer (3)

Sol. $NH_3 \Rightarrow$ sp^3 hybridised with 1 lone pair.

Structure will be Trigonal Pyramidal.

$BrF_5 \Rightarrow$ sp^3d^2 hybridised with 1 lone pair.

Structure will be Square Pyramidal.

$XeF_4 \Rightarrow$ sp^3d^2 with two lone pairs.

Structure will be Square Planar.

$SF_6 \Rightarrow$ sp^3d^2 with no lone pair.

Structure will be Octahedral.

A-I, B-IV, C-II, D-III

67. Match List I with List II.

List I (Complex)	List II (Type of isomerism)
A. $[Co(NH_3)_5(NO_2)]Cl_{(4)}$	I. Solvate isomerism
B. $[Co(NH_3)_5(SO_4)]Br$	II. Linkage isomerism
C. $[Co(NH_3)_6][Cr(CN)_6]$	III. Ionization isomerism
D. $[Co(H_2O)_6]Cl_3$	IV. Coordination isomerism

Choose the correct answer from the options given below:

(1) A-I, B-IV, C-III, D-II
(2) A-II, B-IV, C-III, D-I
(3) A-II, B-III, C-IV, D-I
(4) A-I, B-III, C-IV, D-II

Answer (3)

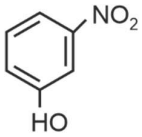
Sol. A. $[Co(NH_3)_5(NO_2)]Cl_{(4)}$ — II. Linkage isomerism due to 'N' and 'O' linkage by NO_2

B. $[Co(NH_3)_5(SO_4)]Br$ — III. Ionization isomerism

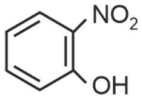
C. $[Co(NH_3)_6][Cr(CN)_6]$ — IV. Coordination isomerism

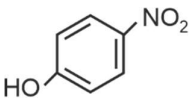
D. $[Co(H_2O)_6]Cl_3$ I.Solvate isomerism

68. Intramolecular hydrogen bonding is present in

(1) 

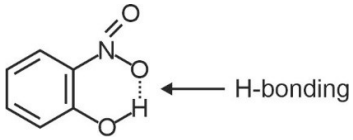
(2) HF

(3) 

(4) 

Answer (3)

Sol. In o-nitrophenol intramolecular H-bonding is present.



69. Given below are two statements:

Statement I : Aniline does not undergo Friedel-Crafts alkylation reaction.

Statement II : Aniline cannot be prepared through Gabriel synthesis.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is correct but Statement II is false
(2) Statement I is incorrect but Statement II is true
(3) Both statement I and Statement II are true
(4) Both Statement I and Statement II are false

Answer (3)

Sol. • Aniline does not undergo Friedel-Crafts alkylation reaction due to salt formation with aluminium chloride, the Lewis acid, which is used as a catalyst.

• Aniline (aromatic primary amine) cannot be prepared by Gabriel phthalimide synthesis because aryl halides do not undergo nucleophilic substitution with anion formed by phthalimide.

70. The highest number of helium atoms is in

(1) 4 g of helium
(2) 2.271098 L of helium at STP
(3) 4 mol of helium
(4) 4 u of helium

Answer (3)

Sol. (1) 4 g of Helium = $\frac{4\text{ g}}{4\text{ g}}$ mole = 1 mole = N_A He atom

(2) 2.2710982 of He at STP = $\frac{2.271}{22.710982}$ mole

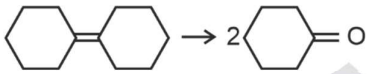
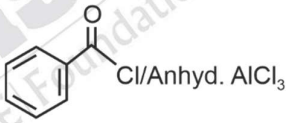
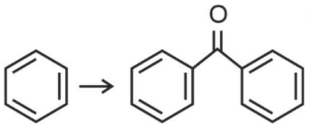
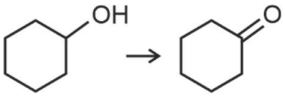
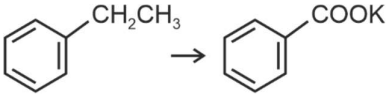
= 0.1 mole

= 0.1 N_A He atom

(3) 4 mol of He = 4 N_A He atoms

(4) 4 u of He = $\frac{4\text{ u}}{4\text{ u}}$ = 1 He atom

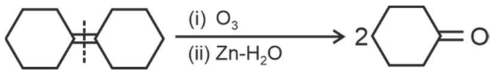
71. Match List I with List II.

List I (Reaction)		**List II (Reagents/Condition)**	
A.		I.	
B.		II.	CrO_3
C.		III.	$KMnO_4/KOH, \Delta$
D.		IV.	(i) O_3 (ii) Zn-H_2O

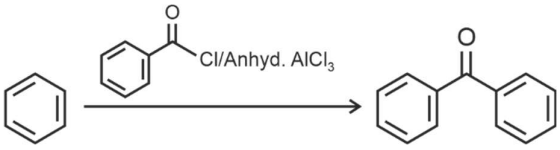
Choose the correct answer from the options given below:

(1) A-IV, B-I, C-II, D-III
(2) A-I, B-IV, C-II, D-III
(3) A-IV, B-I, C-III, D-II
(4) A-III, B-I, C-II, D-IV

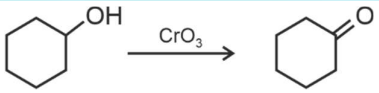
Answer (1)

Sol. (A) 

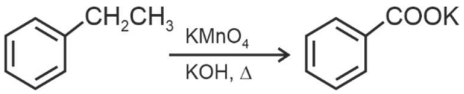
It is reductive ozonolysis

(B) 

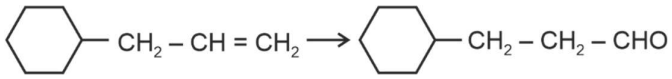
It is Friedel-Crafts acylation reaction.

(C) 

Secondary alcohols are oxidised to ketones by CrO_3

(D) 

72. Identify the correct reagents that would bring about the following transformation.



(1) (i) BH_3

(ii) $H_2O_2 / \overset{\ominus}{O}H$

(iii) alk.$KMnO_4$

(iv) $H_3O^{\oplus}$

(2) (i) H_2O/H^+

(ii) PCC

(3) (i) H_2O/H^+

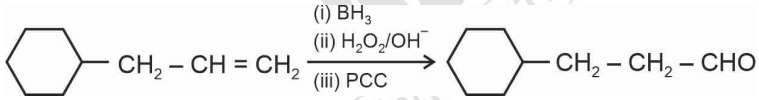
(ii) CrO_3

(4) (i) BH_3

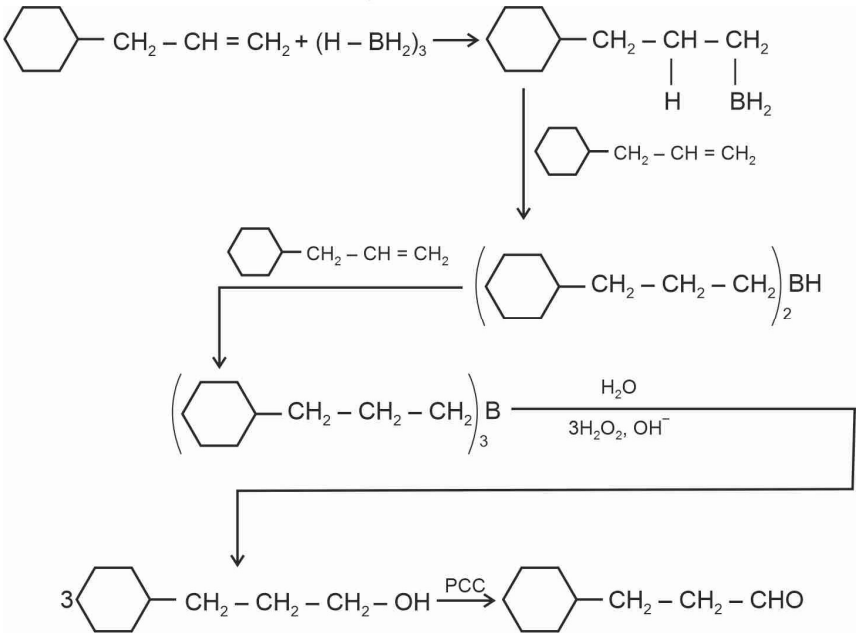
(ii) $H_2O_2 / \overset{\ominus}{O}H$

(iii) PCC

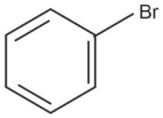
Answer (4)

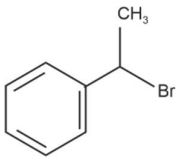
Sol. 

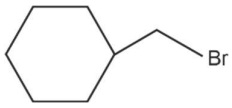
Mechanism :

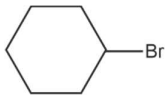


73. The compound that will undergo S_N1 reaction with the fastest rate is

(1) 

(2) 

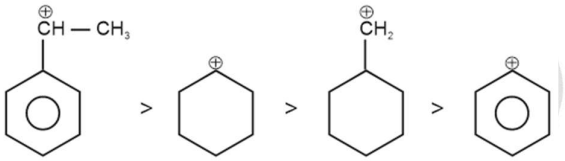
(3) 

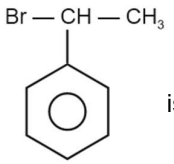
(4) 

Answer (2)

Sol. Reactivity towards S_N1 depends upon stability of carbocation.

Order of stability is



Hence  is most reactive

74. Fehling's solution 'A' is

(1) alkaline solution of sodium potassium tartrate (Rochelle's salt)

(2) aqueous sodium citrate

(3) aqueous copper sulphate

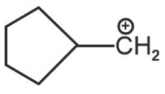
(4) alkaline copper sulphate

Answer (3)

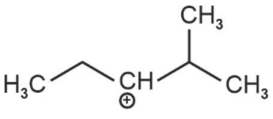
Sol. Fehling solution 'A' = Aqueous copper sulphate

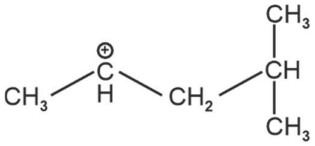
Fehling solution 'B' = Alkaline sodium potassium tartrate (Rochelle salt)

75. The most stable carbocation among the following is :

(1) 

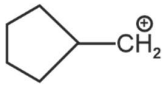
(2) 

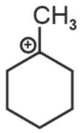
(3) 

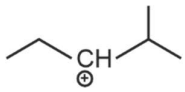
(4) 

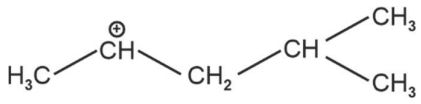
Answer (2)

Sol. The stability of carbocation can be described by the hyperconjugation. Greater the extent of hyperconjugation, more is the stability of carbocation.

(1)  → 1 α-H

(2)  → 7 α-H

(3)  → 3 α-H

(4)  → 5 α-H

Stability order of carbocations = (2) > (4) > (3) > (1)

76. For the reaction $2A \rightleftharpoons B + C$, $K_C = 4 \times 10^{-3}$. At a given time, the composition of reaction mixture is: $[A] = [B] = [C] = 2 \times 10^{-3}$ M.

Then, which of the following is correct?

(1) Reaction has a tendency to go in backward direction.

(2) Reaction has gone to completion in forward direction.

(3) Reaction is at equilibrium.

(4) Reaction has a tendency to go in forward direction.

Answer (1)

Sol. $2A \rightleftharpoons B + C$, $K_C = 4 \times 10^{-3}$

At a given time t, Q_C is to be calculated and been compared with K_C.

$$Q_C = \frac{[B][C]}{[A]^2} = \frac{(2\times10^{-3})(2\times10^{-3})}{(2\times10^{-3})^2}$$

$Q_C = 1$

As $Q_C > K_C$, so reaction has a tendency to move backward.

77. The Henry's law constant (K_H) values of three gases (A, B, C) in water are 145, 2×10^{-5} and 35 kbar, respectively. The solubility of these gases in water follow the order:

(1) A > C > B (2) A > B > C

(3) B > A > C (4) B > C > A

Answer (4)

Sol. Value of Henry's law constant $\propto \frac{1}{\text{Solubility of gas}}$

Higher the value of K_H at a given pressure, lower is the solubility of the gas in the liquid.

K_H value of gases (given) : A > C > B

$\therefore$ Order of solubility of gases in water : B > C > A

78. A compound with a molecular formula of C_6H_{14} has two tertiary carbons. Its IUPAC name is :

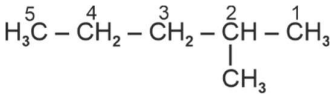
(1) 2,3-dimethylbutane (2) 2,2-dimethylbutane

(3) n-hexane (4) 2-methylpentane

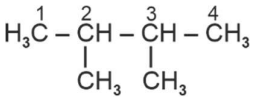
Answer (1)

Sol. $CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - CH_3$ has no tertiary carbon

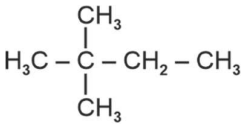
(n-Hexane)


has only one tertiary carbon

(2-Methylpentane)


has two tertiary carbon.

(2, 3-Dimethylbutane)


has no tertiary carbon

(2, 2-Dimethylbutane)

79. In which of the following equilibria, K_p and K_c are **NOT** equal?

(1) $CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)}$

(2) $2BrCl_{(g)} \rightleftharpoons Br_{2(g)} + Cl_{2(g)}$

(3) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

(4) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$

Answer (3)

Sol. $K_p = K_c (RT)^{\Delta n_g}$

for $K_p \neq K_c$,

$\Delta n_g \neq 0$

$\Delta n_g = n_p - n_r$

(1) $\Delta n_g = 2 - 2 = 0$

(2) $\Delta n_g = 2 - 2 = 0$

(3) $\Delta n_g = 2 - 1 = 1$

(4) $\Delta n_g = 2 - 2 = 0$

80. Which reaction is **NOT** a redox reaction?

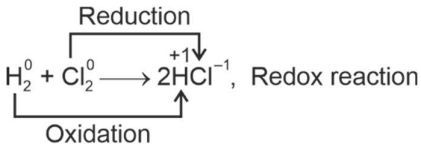
(1) $H_2 + Cl_2 \rightarrow 2HCl$

(2) $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$

(3) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

(4) $2KClO_3 + I_2 \rightarrow 2KIO_3 + Cl_2$

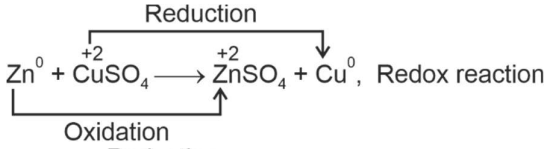
Answer (2)

Sol. (1) 

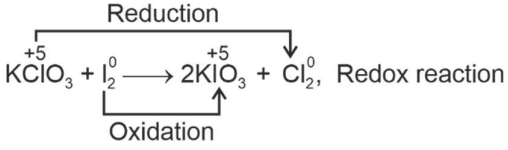
Redox reaction

(2) $\overset{+2}{Ba}\overset{-1}{Cl_2} + \overset{+1}{Na_2}\overset{-2}{SO_4} \longrightarrow \overset{+2}{Ba}SO_4 + 2\overset{+1}{Na}\overset{-1}{Cl}$

This is not a redox reaction as there is no change in oxidation state.

(3) 

Redox reaction

(4) 

Redox reaction

81. Given below are two statements :

Statement I: Both $[Co(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$ complexes are octahedral but differ in their magnetic behaviour.

Statement II: $[Co(NH_3)_6]^{3+}$ is diamagnetic whereas $[CoF_6]^{3-}$ is paramagnetic.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

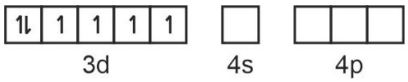
(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

Answer (3)

Sol. In $[Co(NH_3)_6]^{3+}$, Co^{3+} ion is having $3d^6$ configuration.

Electronic configuration of Co^{3+} :



In presence of NH_3 ligand, pairing of electrons takes place and it becomes diamagnetic complex ion.

In presence of NH_3 ligand :

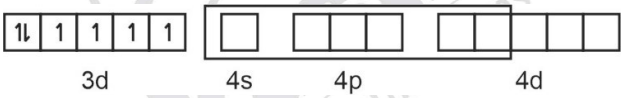


$\therefore$ $[Co(NH_3)_6]^{3+}$ is octahedral with d^2sp^3 hybridisation and it is diamagnetic in nature.

In case of $[CoF_6]^{3-}$, Co is in +3 oxidation state and it is having $3d^6$ configuration.

In presence of weak field F^- ligand, pairing does not take place.

In presence of F^- ligands :



$\therefore$ In $[CoF_6]^{3-}$, Co^{3+} is sp^3d^2 hybridised with four unpaired electrons, so it is paramagnetic in nature.

82. 1 gram of sodium hydroxide was treated with 25 mL of 0.75 M HCl solution, the mass of sodium hydroxide left unreacted is equal to

(1) Zero mg

(2) 200 mg

(3) 750 mg

(4) 250 mg

Answer (4)

Sol. $M = \dfrac{W \times 1000}{M_2 \times V\ (\text{in mL})}$

$W = \dfrac{M \times M_2 \times V\ (\text{in mL})}{1000} = \dfrac{0.75 \times 36.5 \times 25}{1000}$

$= 0.684$ g (Mass of HCl)

$\underset{36.5\,g}{HCl} + \underset{40\,g}{NaOH} \longrightarrow HCl + NaOH$

36.5 g HCl reacts with NaOH = 40 g

0.684 g HCl reacts with $NaOH = \dfrac{40}{36.5} \times 0.684 \simeq 0.750$ g

Amount of NaOH left = 1 g – 0.750 g = 0.250 g = 250 mg

83. Among Group 16 elements, which one does **NOT** show –2 oxidation state?

(1) Te

(2) Po

(3) O

(4) Se

Answer (2)

Sol. Oxygen shows –2, –1, +1 and +2 oxidation states

Selenium shows –2, +2, +4 and +6 oxidation states

Tellurium shows –2, +2, +4 and +6 oxidation states

Polonium shows +2 and +4 oxidation states

84. In which of the following processes entropy increases?

A. A liquid evaporates to vapour.

B. Temperature of a crystalline solid lowered from 130 K to 0 K.

C. $2NaHCO_{3(s)} \rightarrow Na_2CO_{3(s)} + CO_{2(g)} + H_2O_{(g)}$

D. $Cl_{2(g)} \rightarrow 2Cl_{(g)}$

Choose the correct answer from the options given below:

(1) A, C and D

(2) C and D

(3) A and C

(4) A, B and D

Answer (1)

Sol. When a liquid evaporates to vapour entropy increases.

$2NaHCO_{3(s)} \rightarrow Na_2CO_{3(s)} + CO_{2(g)} + H_2O_{(g)}$

Number of gaseous product molecules increases so entropy increases.

$Cl_{2(g)} \rightarrow 2Cl_{(g)}$

1 mole $Cl_{2(g)}$ form 2 mol $Cl_{(g)}$. So entropy increases.

85. Match List I with List II.

	List I (Conversion)		**List II (Number of Faraday required)**
A.	1 mol of H_2O to O_2	I.	3F
B.	1 mol of MnO_4^- to Mn^{2+}	II.	2F
C.	1.5 mol of Ca from molten $CaCl_2$	III.	1F
D.	1 mol of FeO to Fe_2O_3	IV.	5F

Choose the correct answer from the options given below:

(1) A-II, B-III, C-I, D-IV

(2) A-III, B-IV, C-II, D-I

(3) A-II, B-IV, C-I, D-III

(4) A-III, B-IV, C-I, D-II

Answer (3)

Sol. $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$

for 2 mole of H_2O = 4F charge is required

for 1 mole of H_2O = $\frac{4F}{2}$ = 2F required

$\overset{+7}{Mn}O_4^- \rightarrow \overset{+2}{Mn}^{2+}$

for 1 mole MnO_4^- 5F charge is required

$Ca^{2+} \xrightarrow{+2e^-} Ca$

For 1 mole Ca^{2+} ion required = 2F

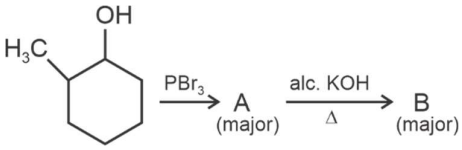
1.5 mole Ca^{2+} ion required = $\frac{2}{1} \times 1.5$ = 3F

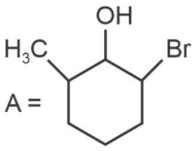
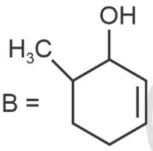
$\overset{+2}{FeO} \rightarrow \overset{+3}{Fe_2O_3}$

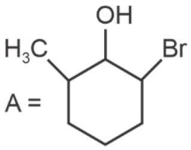
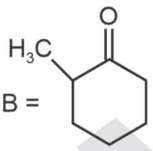
for 1 mole FeO, 1F charge is required.

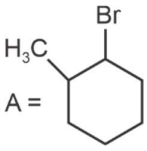
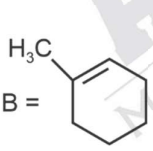
SECTION-B

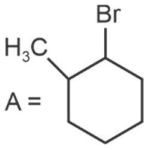
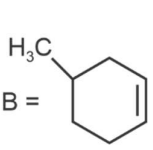
86. Major products A and B formed in the following reaction sequence, are



(1) A =  ; B = 

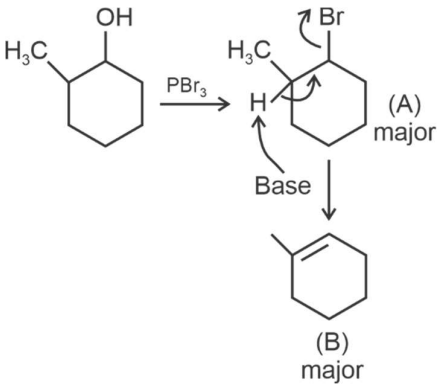
(2) A =  ; B = 

(3) A =  ; B = 

(4) A =  ; B = 

Answer (3)

Sol.



87. Consider the following reaction in a sealed vessel at equilibrium with concentrations of

$N_2 = 3.0 \times 10^{-3}$ M, $O_2 = 4.2 \times 10^{-3}$ M and NO = 2.8×10^{-3} M.

$2NO_{(g)} \rightleftharpoons N_{2(g)} + O_{2(g)}$

If 0.1 mol L^{-1} of $NO_{(g)}$ is taken in a closed vessel, what will be degree of dissociation (α) of $NO_{(g)}$ at equilibrium?

(1) 0.8889

(2) 0.717

(3) 0.00889

(4) 0.0889

Answer (2)

Sol. $2NO_{(g)} \rightleftharpoons N_{2(g)} + O_{2(g)}$

$$K_c = \frac{[N_2][O_2]}{[NO]^2}$$

$$= \frac{3\times10^{-3}\times4.2\times10^{-3}}{2.8\times10^{-3}\times2.8\times10^{-3}}$$

$$= 1.607$$

	$2NO_{(g)}$	$\rightleftharpoons$	$N_{2(g)}$	+	$O_{2(g)}$
t = 0	0.(3)		0		0
	$0.1 - 0.1\alpha$		0.05α		0.05α

$$K_c = \frac{0.05\alpha\times0.05\alpha}{(0.1-0.1\alpha)^2}$$

$$K_c = \frac{0.05\alpha\times0.05\alpha}{0.01(1-\alpha)^2}$$

$$1.607 = \frac{(0.05)^2\alpha^2}{0.01(1-\alpha)^2}$$

$$\frac{\alpha^2}{(1-\alpha)^2} = \frac{1.607\times(0.1)^2}{(0.05)^2}$$

$$\frac{\alpha}{1-\alpha} = \frac{1.27\times0.1}{0.05}$$

$$\frac{\alpha}{1-\alpha} = 2.54$$

$$\alpha = 2.54 - 2.54\alpha$$

$$3.54\alpha = 2.54$$

$$\alpha = \frac{2.54}{3.54} = 0.717$$

88. The products A and B obtained in the following reactions, respectively, are

$3ROH + PCl_3 \rightarrow 3RCl + A$

$ROH + PCl_5 \rightarrow RCl + HCl + B$

(1) H_3PO_4 and $POCl_3$

(2) H_3PO_3 and $POCl_3$

(3) $POCl_3$ and H_3PO_3

(4) $POCl_3$ and H_3PO_4

Answer (2)

Sol. These reactions are preparation of haloalkanes from alcohols.

$$3ROH + PCl_3 \longrightarrow 3RCl + \underset{(A)}{H_3PO_3}$$

$$ROH + PCl_5 \longrightarrow RCl + HCl + \underset{(B)}{POCl_3}$$

A and B are H_3PO_3 and $POCl_3$ respectively.

89. Given below are certain cations. Using inorganic qualitative analysis, arrange them in increasing group number from 0 to VI.

A. Al^{3+} B. Cu^{2+}

C. Ba^{2+} D. Co^{2+}

E. Mg^{2+}

Choose the correct answer from the options given below.

(1) E, C, D, B, A (2) E, A, B, C, D

(3) B, A, D, C, E (4) B, C, A, D, E

Answer (3)

Sol.

Group	Cations
Group-II	Cu^{2+}
Group-III	Al^{3+}
Group-IV	Co^{2+}
Group-V	Ba^{2+}
Group-VI	Mg^{2+}

The correct order of group number of ions is $\underset{(B)}{Cu^{2+}} < \underset{(A)}{Al^{3+}} < \underset{(D)}{Co^{2+}} < \underset{(C)}{Ba^{2+}} < \underset{(E)}{Mg^{2+}}$

$\therefore$ The correct order is B, A, D, C, E

90. During the preparation of Mohr's salt solution (Ferrous ammonium sulphate), which of the following acid is added to prevent hydrolysis of Fe^{2+} ion?

(1) dilute nitric acid

(2) dilute sulphuric acid

(3) dilute hydrochloric acid

(4) concentrated sulphuric acid

Answer (2)

Sol. During the preparation of Mohr's salt, dilute sulphuric acid is added to prevent the hydrolysis of Fe^{2+} ion.

91. The work done during reversible isothermal expansion of one mole of hydrogen gas at 25°C from pressure of 20 atmosphere to 10 atmosphere is

(Given R = 2.0 cal K^{-1} mol^{-1})

(1) 413.14 calories

(2) 100 calories

(3) 0 calorie

(4) –413.14 calories

Answer (4)

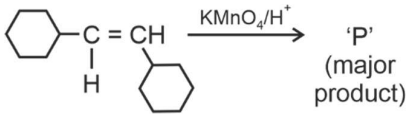
Sol. $W_{rev,\ iso} = -2.303\ nRT \log \frac{P_i}{P_f}$

$= -2.303 \times 1 \times 2 \times 298 \times \log 2$

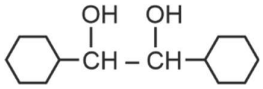
$= -2.303 \times 1 \times 2 \times 298 \times 0.3$

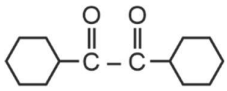
$= -413.14$ calories

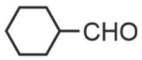
92. For the given reaction:

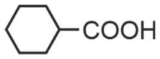


'P' is

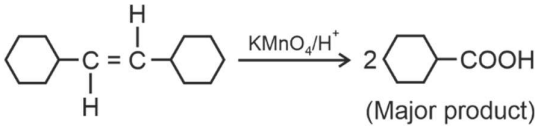
(1) 

(2) 

(3) 

(4) 

Answer (4)

Sol. 

93. The plot of osmotic pressure (Π) vs concentration (mol L^{-1}) for a solution gives a straight line with slope 25.73 L bar mol^{-1}. The temperature at which the osmotic pressure measurement is done is

(Use R = 0.083 L bar mol^{-1} K^{-1})

(1) 25.73°C

(2) 12.05°C

(3) 37°C

(4) 310°C

Answer (3)

Sol. $\Pi = CRT$

Slope = RT

$25.73 = 0.083 \times T$

$$T = \frac{25.73}{0.083} = 309.47 \approx 310 \text{ K}$$

$\therefore$ Temperature in °C = 310 – 273

= 37°C

94. Identify the major product C formed in the following reaction sequence:

$$CH_3 - CH_2 - CH_2 - I \xrightarrow{NaCN} A \xrightarrow[\text{Partial hydrolysis}]{OH^-} B \xrightarrow[Br_2]{NaOH} \underset{\text{(major)}}{C}$$

(1) butanamide

(2) α–bromobutanoic acid

(3) propylamine

(4) butylamine

Answer (3)

Sol.

$$CH_3CH_2CH_2 - I \xrightarrow{NaCN} \underset{(A)}{CH_3CH_2CH_2 - CN} \xrightarrow{\text{Partial hydrolysis}} \underset{(B)}{CH_3CH_2 - CH_2 - \overset{\overset{O}{\|}}{C} - NH_2} \xrightarrow{NaOH/Br_2} \underset{\substack{(C)\\ \text{(Propyl amine)}}}{CH_3CH_2CH_2 - NH_2}$$

- Step-I is S_N reaction with $\overset{\ominus}{C}N$ nucleophile.
- Step-II will give amide.
- Step-III is Hoffmann bromamide degradation reaction.

95. A compound X contains 32% of A, 20% of B and remaining percentage of C. Then, the empirical formula of X is :

(Given atomic masses of A = 64; B = 40; C = 32 u)

(1) AB_2C_2

(2) ABC_4

(3) A_2BC_2

(4) ABC_3

Answer (4)

Sol.

Element	Mass percentage %	No. of moles	No. of moles/ Smallest number	Simplest whole number
A	32%	$\frac{32}{64}=\frac{1}{2}$	$\frac{1}{2}\times 2$	= 1
B	20%	$\frac{20}{40}=\frac{1}{2}$	$\frac{1}{2}\times 2$	= 1
C	48%	$\frac{48}{32}=\frac{3}{2}$	$\frac{3}{2}\times 2$	= 3

So, empirical formula of $X = \begin{matrix} A & : & B & : & C \\ 1 & : & 1 & : & 3 \end{matrix}$

$\therefore$ The correct empirical formula of compound X is ABC_3

96. The rate of a reaction quadruples when temperature changes from 27°C to 57°C. Calculate the energy of activation.

Given R = 8.314 J K^{-1} mol^{-1}, log4 = 0.6021

(1) 3.80 kJ/mol

(2) 3804 kJ/mol

(3) 38.04 kJ/mol

(4) 380.4 kJ/mol

Answer (3)

Sol. $\log\left(\frac{k_2}{k_1}\right)=\frac{E_a}{2.303R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)$

$\log\left(\frac{4}{1}\right)=\frac{E_a}{2.303R}\left(\frac{1}{300}-\frac{1}{330}\right)$

$E_a = \frac{(\log(4))\times 2.303\times 8.314\times 300\times 330}{30}$

$= 3.804 \times 10^4$ J/mol

= 38.04 kJ/mol

97. Identify the **correct** answer.

(1) Dipole moment of NF_3 is greater than that of NH_3

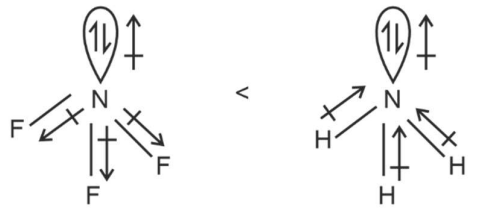
(2) Three canonical forms can be drawn for CO_3^{2-} ion

(3) Three resonance structures can be drawn for ozone

(4) BF_3 has non-zero dipole moment

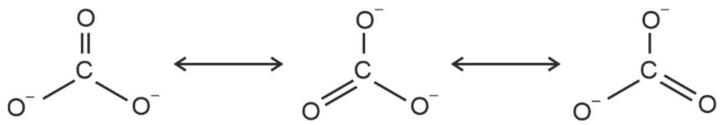
Answer (2)

Sol. (1)



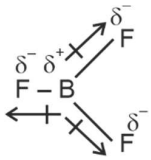
$(0.80 \times 10^{-30}$ Cm) $(4.90 \times 10^{-30}$ Cm)

(2)



(3 canonical forms)

(3) In ozone; there are two resonating structures.

(4) BF_3 i.e.,  ; Dipole moment = 0

98. The pair of lanthanoid ions which are diamagnetic is

(1) Gd^{3+} and Eu^{3+}

(2) Pm^{3+} and Sm^{3+}

(3) Ce^{4+} and Yb^{2+}

(4) Ce^{3+} and Eu^{2+}

Answer (3)

Sol. Magnetic moment $\mu = \sqrt{n(n+2)}$

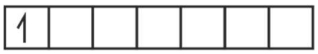
$n \rightarrow$ number of unpaired electron

$Ce^{4+} \Rightarrow (Xe)\ 4f^0$

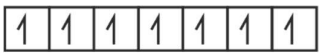
$\mu = 0$ Diamagnetic

$Yb^{2+} \Rightarrow (Xe)\ 4f^{14}$ 

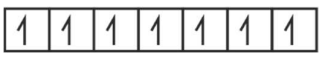
$\mu = 0$ Diamagnetic

$Ce^{3+} \Rightarrow (Xe)\ 4f^{(3)}$ 

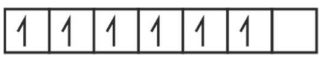
$\mu = \sqrt{3}$ Paramagnetic

$Eu^{2+} \Rightarrow (Xe)\ 4f^7$ 

$\mu = \sqrt{63}$ Paramagnetic

$Gd^{3+} \Rightarrow (Xe)\ 4f^7$ 

$\mu = \sqrt{63}$ Paramagnetic

$Eu^{3+} \Rightarrow (Xe)\ 4f^6$ 

$\mu = \sqrt{48}$ Paramagnetic

$Pm^{3+} \Rightarrow (Xe)\ 4f^4$ | 1 | 1 | 1 | 1 | | | |

$\mu = \sqrt{24}$ Paramagnetic

$Sm^{3+} \Rightarrow (Xe)\ 4f^5$ | 1 | 1 | 1 | 1 | 1 | | |

$\mu = \sqrt{35}$ Paramagnetic

Hence Ce^{4+} and Yb^{2+} are only diamagnetic.

99. Given below are two statements :

Statement I : $[Co(NH_3)_6]^{3+}$ is a homoleptic complex whereas $[Co(NH_3)_4Cl_2]^{+}$ is a heteroleptic complex.

Statement II : Complex $[Co(NH_3)_6]^{3+}$ has only one kind of ligands but $[Co(NH_3)_4Cl_2]^{+}$ has more than one kind of ligands.

In the light of the above statements, choose the *correct* answer from the options given below.

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

Answer (3)

Sol. $[Co(NH_3)_6]^{3+}$ is a homoleptic complex as only one type of ligands (NH_3) is coordinated with Co^{3+} ion. While $[Co(NH_3)_4Cl_2]^{+}$ is a heteroleptic complex in which Co^{3+} ion is ligated with more than one type of ligands, *i.e.*, NH_3 and Cl^-.

100. Mass in grams of copper deposited by passing 9.6487 A current through a voltmeter containing copper sulphate solution for 100 seconds is (Given : Molar mass of Cu : 63 g mol^{-1}, 1 F = 96487 C)

(1) 31.5 g

(2) 0.0315 g

(3) 3.15 g

(4) 0.315 g

Answer (4)

Sol. $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

$$\text{Mass of Cu deposited (w)} = \frac{M \times i \times t}{nF}$$

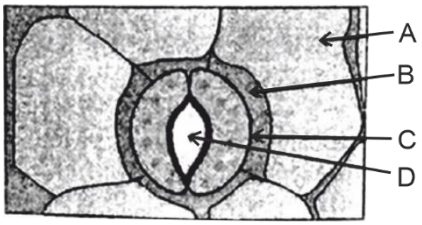
$$= \frac{63 \times 9.6487 \times 100}{2 \times 96487}$$

$$= 0.315 \text{ g}$$

BOTANY

SECTION-A

101. In the given figure, which component has thin outer walls and highly thickened inner walls?



(1) A (2) B

(3) C (4) D

Answer (3)

Sol. Guard cells of stomata have thin outer wall and highly thickened inner walls.

102. List of endangered species was released by

(1) FOAM (2) IUCN

(3) GEAC (4) WWF

Answer (2)

Sol. List of endangered species was released by – IUCN.

103. The type of conservation in which the threatened species are taken out from their natural habitat and placed in special setting where they can be protected and given special care is called

(1) Semi-conservative method (2) Sustainable development

(3) *in-situ* conservation (4) Biodiversity conservation

Answer (4)

Sol. The type of conservation in which threatened species are taken out from their natural habitat and placed in special setting where they can be protected and given special care is called *ex-situ* conservation which is a type of biodiversity conservation.

104. Which one of the following can be explained on the basis of Mendel's Law of Dominance?

A. Out of one pair of factors one is dominant and the other is recessive.

B. Alleles do not show any expression and both the characters appear as such in F_2 generation.

C. Factors occur in pairs in normal diploid plants.

D. The discrete unit controlling a particular character is called factor.

E. The expression of only one of the parental characters is found in a monohybrid cross.

Choose the correct answer from the options given below:

(1) B, C and D only (2) A, B, C, D and E

(3) A, B and C only (4) A, C, D and E only

Answer (4)